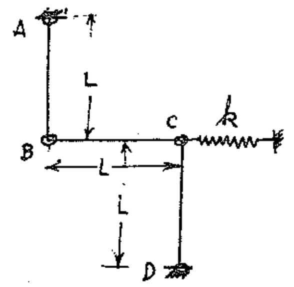
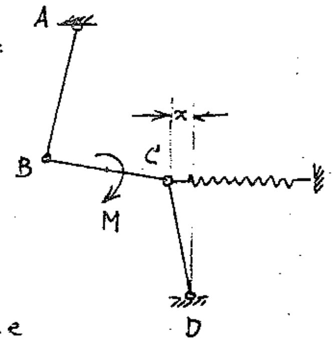
9.3 (a)

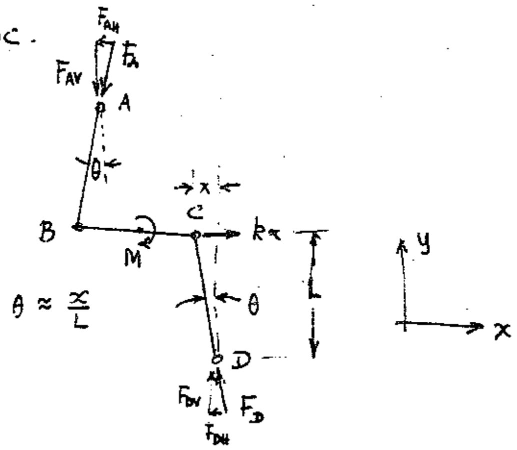


Displaced:



Since B, C can transmit no moments, the reactions at A, D must be along AB, DC.

Consider a free-body:



Equilibrium:

$\sum M_D = 0 : (F_{AV} + 2F_{AH})L = M + kxL$ ---- (1)

$\sum F_y = 0 : \quad F_{AV} = F_{DV}$ ---- (2)

$\sum F_z = 0 : \quad F_{AH} + F_{DH} = kx$ ---- (3)

From geometry, $F_{AH} \approx F_{AV}\frac{x}{L}$ -- (4)

$F_{DH} \approx F_{DV}\frac{x}{L}$ -- (5)

From these, $M = \frac{kL^2}{2}$

(b) When M is reversed, there is no problem of stability. The reversed M can be increased indefinitely or until (i) AD or CD fails in tension
(ii) BC fails in bending.

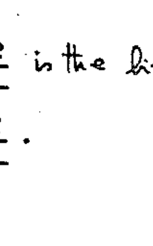
9.4

For Equilibrium: $\sum M_{Bz} = 0$

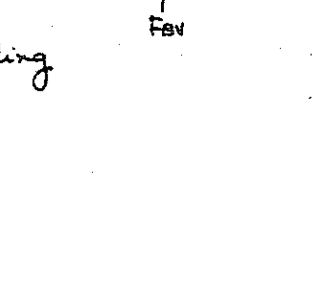
$P(L\sin\theta + \epsilon) = k\theta$

For small θ, $\sin\theta \approx \theta$

$\therefore \quad \theta = \frac{P\epsilon}{k - PL}$

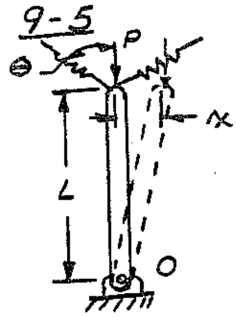


Free-Body



As $P \to \frac{k}{L}$, $\theta \to \infty$. Therefore $P = \frac{k}{L}$ is the limiting load. Equilibrium is stable for $P < \frac{k}{L}$.

9-5



For equilibrium in the deflected position

$\circlearrowleft \Sigma M_o = 0$

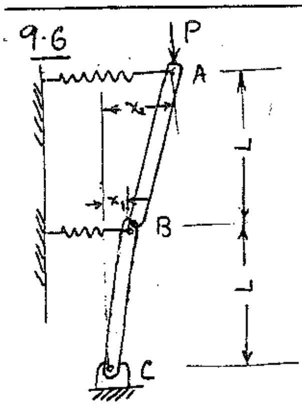
$$-P\alpha + 2k\,\delta L \sin\theta = 0$$

Where δ is the spring deflection

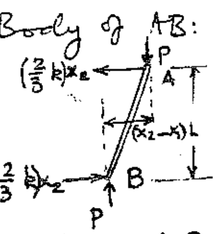
$$\delta = \alpha \sin\theta$$

$$P_{CRIT} = 2kL\sin^2\theta$$

9.6

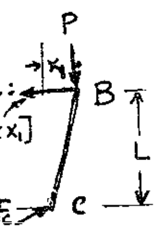


Free Body of AB:



$\Sigma M_B = 0: \quad P(x_2 - x_1) = \frac{2}{3}kx_2L$ —— (1)

Free-Body of BC: $[\frac{2}{3}kx_2 + kx_1]$



$\Sigma M_c = 0: \quad (P)(x_1) = k(x_1 + \frac{2}{3}x_2)L$ —— (2)

9-3

9.6 continued.

Equilibrium is possible only if (i) and (ii) have a solution. To determine the condition for this, eliminate x_1 from equation (2) to obtain

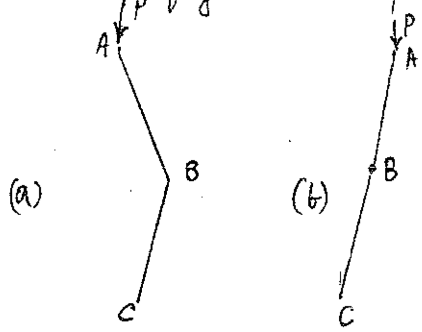
$$\left[\left(2-\frac{kL}{P}\right)\left(1-\frac{2}{3}\frac{kL}{P}\right)-1\right]x_2 = 0.$$

If $x_2 = 0$, we get $x_1 = 0$ and the solution is trivial. The alternative is for x_1, x_2 to be arbitrary and

$$\left[\left(2-\frac{kL}{P}\right)\left(1-\frac{2}{3}\frac{kL}{P}\right)-1\right] = 0.$$

This gives $P = \dfrac{kL}{3}$ or $P = 2kL$

Stability: Neutral Equilibrium is possible with $P = \dfrac{kL}{3}$ if $x_2 = -x_1$, fig (a). Also possible with $P = 2kL$ if $x_2 = \frac{3}{2}x_1$, fig (b)



If we examine stability by writing restoring moments, we find:

(a) is stable when $P < \dfrac{kL}{3}$

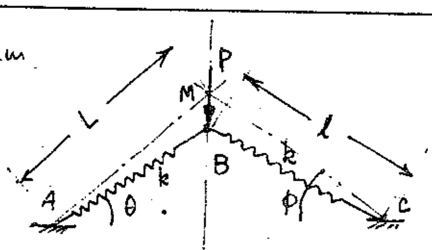
unstable when $P > \dfrac{kL}{3}$

(b) is stable when $P < 2kL$

unstable when $P > 2kL$

The structure as a whole is stable only when all possible configurations are stable, i.e., when $P < \dfrac{kL}{3}$. The neutral equilibrium case (b) is of only academic interest.

9.7 From a free-body diagram, equilibrium requirements give



$2k(AM-AB)\sin\theta = P$

From geometry, $AM\cos\phi = AB\cos\theta$

For small θ, ϕ, write $\sin\theta \approx \theta$

$\cos\phi \approx 1 - \dfrac{\phi^2}{2}$

$\cos\theta \approx 1 - \dfrac{\theta^2}{2}$ Then $P = kL\theta(\phi^2-\theta^2)$ —(1)

9.7 continued

$$\frac{dP}{d\theta} = kL(-\phi^2 + 3\theta^2) = 0 \quad \text{for snap through.}$$

This happens at $\theta = \pm \frac{\phi}{\sqrt{3}}$. (both above and below AC. Sign of P changes for snap-through below)

at $P = \pm \frac{2kL\phi^3}{3\sqrt{3}}$, snap through occurs.

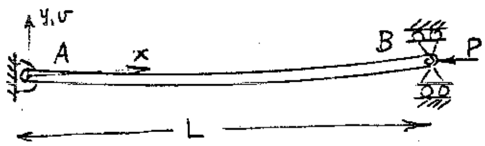
Stability: If during a perturbation through an angle $d\theta = \epsilon$ the force P is constant, the restoring force is

$$P_{rest.} = kL(\theta+\epsilon)\left[-\phi^2 + (\theta+\epsilon)^2\right] - P \approx kL\epsilon(-\phi^2 + 3\theta^2)$$

Therefore equilibrium is stable for $\phi^2 < 3\theta^2$, ie, $|\theta| > \frac{\phi}{\sqrt{3}}$

unstable for $|\theta| \lesssim \frac{\phi}{\sqrt{3}}$

9.8 The differential equation 9.5 applies.



$$EI\frac{d^2v}{dx^2} = M_b = -Pv \quad \text{--- (1)}$$

The general solution is $v = A\cos Kx + B\sin Kx \qquad K \equiv \sqrt{\frac{P}{EI}}$

Boundary Conditions: $v = 0$ at $x = 0$, $x = L$

These give $A = 0$, $B\sin KL = 0$

Therefore, for a non-trivial solution, B is indeterminate and

$$\sin KL = 0 \quad \text{———— (2)}$$

The solutions to (2) are $kL = n\pi$, $n = 1, 2, 3 \cdots$

Thus the configurations $v_n = B_n \sin n\pi\frac{x}{L}$ are neutral equilibrium configurations for compressive loads $P_n = \frac{n^2\pi^2 EI}{L^2}$.

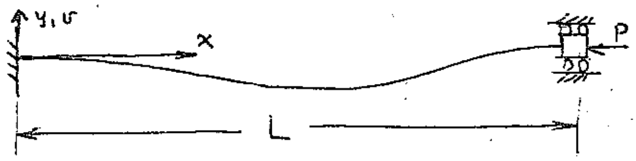
If we examine stability, configuration v_n is:

stable for $P < P_n$

unstable for $P > P_n$.

Structure as a whole is stable only when all possible configurations are stable, ie, when $P < P_1 = \frac{\pi^2 EI}{L^2}$

9.9 Equation 9.10 applies:



$$v = C_1 + C_2 x + C_3 \sin x\sqrt{\frac{P}{EI}} + C_4 \cos x\sqrt{\frac{P}{EI}}$$

Boundary Conditions: $v = \frac{dv}{dx} = 0$ at $x = 0$, $x = L$.

Using these, get

$$\begin{bmatrix} C_1 + C_4 = 0 \\ C_2 + C_3\sqrt{\frac{P}{EI}} = 0 \\ C_1 + C_2 L + C_3 \sin L\sqrt{\frac{P}{EI}} + C_4 \cos L\sqrt{\frac{P}{EI}} = 0 \\ C_2 + C_3\sqrt{\frac{P}{EI}}\cos L\sqrt{\frac{P}{EI}} - C_4\sqrt{\frac{P}{EI}}\sin L\sqrt{\frac{P}{EI}} = 0 \end{bmatrix}$$

Eliminating C_2, C_4

$$C_1\left(1 - \cos L\sqrt{\frac{P}{EI}}\right) + C_3\left(\sin L\sqrt{\frac{P}{EI}} - L\sqrt{\frac{P}{EI}}\right) = 0$$

$$C_1\sqrt{\frac{P}{EI}}\sin L\sqrt{\frac{P}{EI}} + C_3\sqrt{\frac{P}{EI}}\left(\cos L\sqrt{\frac{P}{EI}} - 1\right) = 0$$

For a non-trivial solution, determinant of coefficients of C_1, C_3 must vanish. Thus

$$(1-\cos\beta L)(\cos\beta L - 1) - \sin^2\beta L + \beta L \sin\beta L = 0, \quad \beta \equiv \sqrt{\frac{P}{EI}}$$

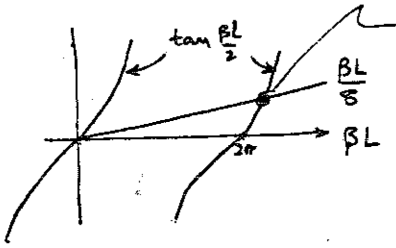
This may be reduced to

$$\sin\frac{\beta L}{2}\left(\frac{\beta L}{2}\cos\frac{\beta L}{2} - 4\sin\frac{\beta L}{2}\right) = 0$$

Roots: One family (symmetric modes) $\frac{\beta L}{2} = n\pi$, $n = 1, 2, 3 \cdots$

giving $P_n = \frac{EI(2n)^2\pi^2}{L^2}$, $v_n = C_1\left(1 - \cos\frac{2n\pi x}{L}\right)$

Other family (antisymmetric modes): $\frac{\beta L}{8} = \tan\frac{\beta L}{2}$



Lowest root is greater than $n=1$ root of other family.

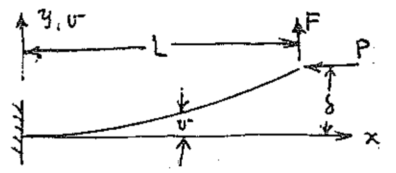
Thus structure is stable only for

$$P < P_{cr} = \frac{4EI\pi^2}{L^2}$$

9.10

Eqn 9.9 applies:

$$EI\frac{d^4v}{dx^4} + P\frac{d^2v}{dx^2} = 0.$$



Solution is 9.10:

$$v = C_1 + C_2 x + C_3 \sin x\sqrt{\frac{P}{EI}} + C_4 \cos x\sqrt{\frac{P}{EI}}$$

Boundary Conditions : at $x=0$ $v = \frac{dv}{dx} = 0$

at $x=L$ shear force $= V_y = EI\frac{d^3v}{dx^3} = 0$ $\quad M_b = EI\frac{d^2v}{dx^2} = 0$

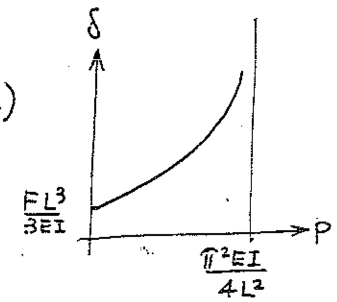
These give $C_1 = -C_4 = \delta + \frac{F}{P}L$

$C_3 = -C_2\sqrt{\frac{EI}{P}} = (\delta + \frac{F}{P}L)\cot KL$ $\quad K \equiv \sqrt{\frac{P}{EI}}$

and $v = (\delta + \frac{F}{P}L)(1-\cos Kx) - (\delta + \frac{F}{P}L)\cot KL(xK - \sin Kx)$

$\delta = (v)_{x=L} = (\delta + \frac{F}{P}L)(1 - LK\cot KL)$

This gives $\delta = \frac{F}{P}L\frac{(\sin KL - LK\cos KL)}{\cos KL}$



9.11 Eqn (9.2.1) may be written

$$\frac{P}{AE} \geq \frac{Y}{E} \text{ for plastic flow } \text{———} (9.25a)$$

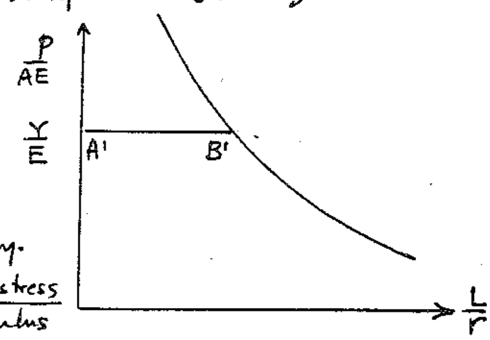
Eqn (9.26) may be written

$$\frac{P}{AE} < \frac{\pi^2}{4}\frac{I}{AL^2} = \frac{\pi^2}{4}\frac{r^2}{L^2} \text{ for stability, or}$$

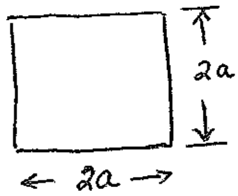
$$\frac{P}{AE} \geq \frac{\pi^2}{4}\left(\frac{r}{L}\right)^2 \text{ for instability } \text{——} (9.26a)$$

Equation (9.26a) describes the instability locus for cantilever columns. We see that in a plot of $\frac{P}{AE}$ versus $\left(\frac{L}{r}\right)^2$ it is a fixed curve, independent of material & geometry.

The ordinate corresponding to AB is $\frac{Y}{E} = \frac{\text{yield stress}}{\text{modulus}}$



9.12



FAILURE BY YIELDING $P = AY = 4a^2 Y$

For 1020 HR, $Y \approx 315\ MN/m^2$, (FIG 5.5a)

$P = 50\ kN$

$\therefore\ a = \sqrt{\dfrac{50}{4 \times 315 \times 10^3}} = 63\ mm$

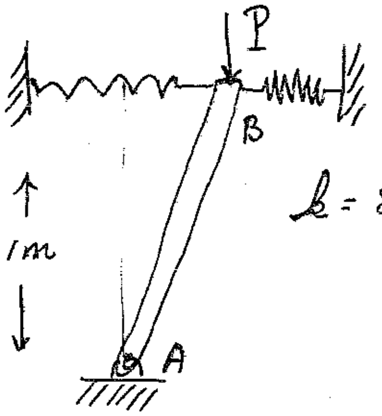
BUCKLING FAILURE

$$P = \frac{\pi^2}{4}\frac{EI}{L^2} \ :\ I = \frac{1}{2} b h^3 = \frac{4}{3} a^4$$

$$\therefore P = \frac{\pi^2}{3}\frac{E a^4}{L^2} \ :\ a^4 = \frac{3L^2 P}{\pi^2 E} = 23.4\ mm$$

$\therefore$ WE MUST HAVE $a > 23.4\ mm$ to support a LOAD $> 50\ kN$

9.13



$k = 80\ kN/m$

FOR BUCKLING MODE SHOWN, TAKE MOMENTS ABOUT A

$P = 1 \times 2 \times 80 = 160\ kN$

P_{CRIT} FOR YIELDING $= YA = 385 \times 491$

$= 189\ kN$

ANOTHER BUCKLING MODE IS THAT B DOES NOT MOVE SIDEWAYS INSTEAD AB BEHAVES AS A HINGED COLUMN.

$$P_{CRIT} = \frac{\pi^2 EI}{L^2} = \pi^2 \times 75 \times 10^6 \times \frac{\pi}{64} \times 3.9 \times 10^{-7}$$

$= 14.1\ kN$ HENCE THIS IS FAILURE MODE

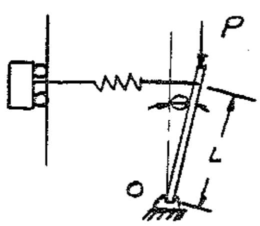
9.14 Equation 9.17 may be written

$$\delta = \frac{M_o}{P}\left(\frac{1-\cos L\sqrt{\frac{P}{EI}}}{\cos L\sqrt{\frac{P}{EI}}}\right)$$

For small values of $\theta = L\sqrt{\frac{P}{EI}}$, $1-\cos\theta \approx \frac{\theta^2}{2}$,
$\cos\theta \approx 1.0$

$$\therefore\ \delta \approx \frac{M_o}{P}\frac{\theta^2}{2} = \frac{M_o L^2}{2EI}$$

9-15



$+\circlearrowright \Sigma M_o = 0$

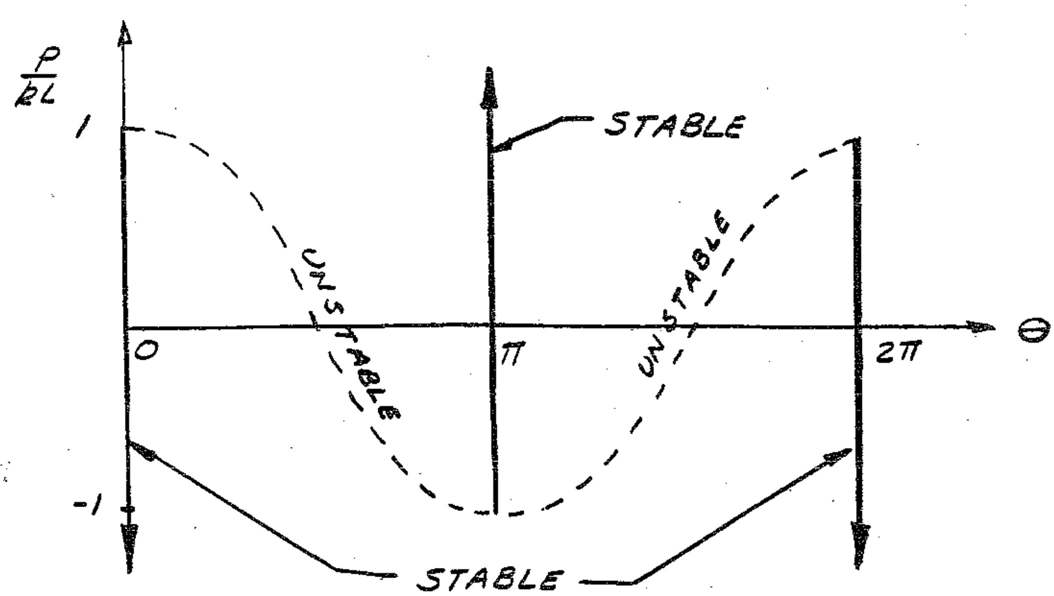
$$-PL\sin\theta + kL^2\sin\theta\cos\theta = 0$$
$$PL\sin\theta = kL^2\sin\theta\cos\theta$$

a) $P_{CRIT} = kL$

b) $P = kL\cos\theta$ FOR $\begin{cases} 0 < \theta < \pi \\ \pi < \theta < 2\pi \end{cases}$

ALSO $P < kL$ FOR $\theta = 0$
$P > -kL$ FOR $\theta = \pi$

c)

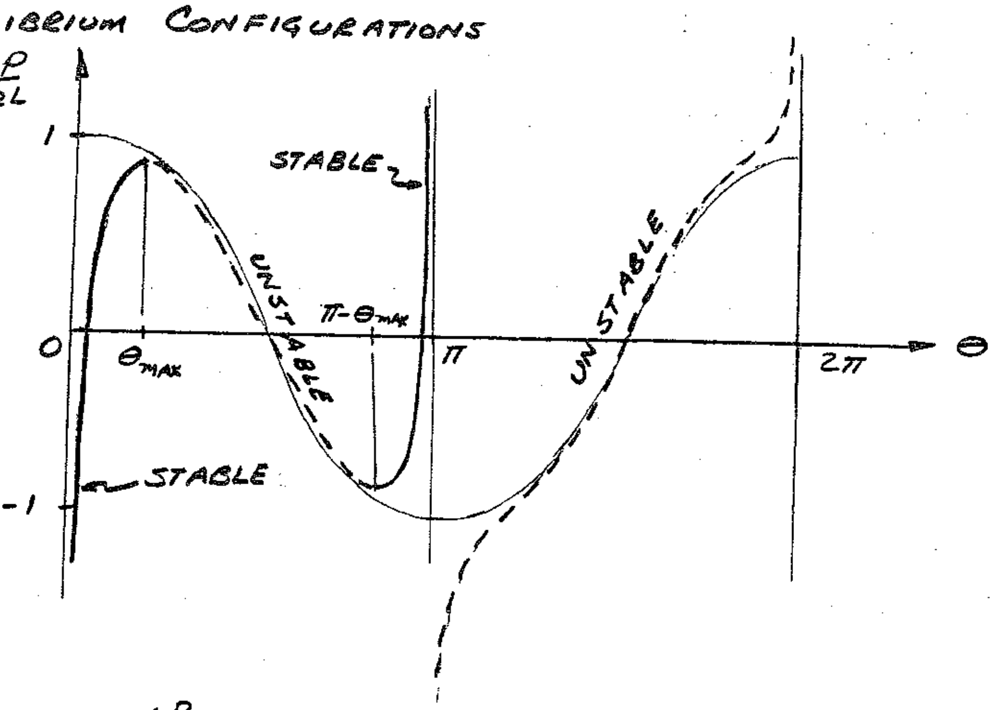


d) EQUILIBRIUM $kL^2\cos\theta(\sin\theta - \sin\epsilon) = PL\sin\theta$

$$\frac{P}{kL} = \frac{\sin\theta - \sin\epsilon}{\tan\theta}$$

9.15 cont

e) EQUILIBRIUM CONFIGURATIONS



AT θ_{max} $\frac{d\left(\frac{P}{kL}\right)}{d\theta} = 0$

$$\sin^3 \theta_{max} = \sin \epsilon$$

f) SNAP THROUGH OCCURS AT $\theta = \theta_{max}$

WITH $\frac{P_{max}}{kL} = \left[1 - (\sin\epsilon)^{2/3}\right]^{3/2}$

g) $\theta_1 = \theta_{max}$

h) FOR SMALL θ, $\cos\theta \approx 1 - \frac{\theta^2}{2}$

$\sin\theta \approx \theta$

$L\theta \rightarrow x$

FROM PART a

$$kL^2\theta\left(1 - \frac{\theta^2}{2}\right) = PL\theta$$

$$kLx\left(1 - \frac{1}{2}\frac{x^2}{L^2}\right) = Px$$

$$\beta = \frac{-1}{2}$$

9.16 Equation 5.29 gives

$$\bar{\sigma} = \left\{ \frac{1}{2}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right]\right\}^{\frac{1}{2}}$$

For a spherical shell under biaxial tension, $\sigma_1 = \sigma_2$, $\sigma_3 = 0$

By considering equilibrium of a hemispherical free-body, obtain

$$p = \frac{2\sigma_1 t_0}{r_0} \quad \text{where } p = \text{pressure} \qquad \text{-----} \quad (1)$$

From above, $\bar{\sigma} = \sigma_1 = \sigma_2$

$$\therefore \quad p = \frac{2\bar{\sigma} t_0}{r_0} \qquad \text{-------------} \quad (2)$$

Equation 5.30 gives

$$\bar{\epsilon} = \int\left\{\frac{2}{9}\left[(d\epsilon_I-d\epsilon_{II})^2+(d\epsilon_{II}-d\epsilon_{III})^2+(d\epsilon_{III}-d\epsilon_I)^2\right]\right\}^{\frac{1}{2}}$$

The volume increase (dilatation) of an infinitesimal element of volume during an infinitesimal increase in strain is

$$d(\Delta V) = (d\epsilon_I + d\epsilon_{II} + d\epsilon_{III})\Delta V$$

For no increase in volume, $d\epsilon_I + d\epsilon_{II} + d\epsilon_{III} = 0$.

For a spherical shell, $d\epsilon_I = d\epsilon_{II} = \frac{dr_0}{r_0}$ $\quad \therefore \; d\epsilon_{III} = -2\frac{dr_0}{r_0}$

ie, $\frac{dt_0}{t_0} = -2\frac{dr_0}{r_0}$

$$\therefore \quad \bar{\epsilon} = \int \frac{2}{3}\frac{dr_0}{r_0} \quad \text{or } d\bar{\epsilon} = 2\frac{dr_0}{r_0}$$

Differentiating the logarithm of (2),

$$\frac{dp}{p} = \frac{d\bar{\sigma}}{\bar{\sigma}} + \frac{dt_0}{t_0} - \frac{dr_0}{r_0} = \frac{d\bar{\sigma}}{\bar{\sigma}} - \frac{3}{2}d\bar{\epsilon} \qquad \text{-----------} \quad (3)$$

9.15 continued

Initially, $\frac{d\bar{\sigma}}{\bar{\sigma}} - \frac{3}{2}\,d\bar{\epsilon} > 0$ ie,

$$\frac{d\bar{\sigma}}{d\bar{\epsilon}} > \frac{3}{2}\bar{\sigma} \quad \text{because} \quad \frac{d\bar{\sigma}}{d\bar{\epsilon}} = E \text{ initially}$$

$\therefore$ Initially $dp > 0$, or p increases.

When $\frac{d\bar{\sigma}}{d\bar{\epsilon}} = \frac{3}{2}\bar{\sigma}$, $dp = 0$

$\frac{d\bar{\sigma}}{d\bar{\epsilon}} < \frac{3}{2}\bar{\sigma}$, $dp < 0$, p decreases

$\therefore$ The max. pressure occurs at $\frac{d\bar{\sigma}}{d\bar{\epsilon}} = \frac{3}{2}\bar{\sigma}$.

Integrating (3),

$$\ln p = \ln\bar{\sigma} - \frac{3}{2}\bar{\epsilon} + \underbrace{\ln C_1}_{\text{Constant of Integration}}$$

or $\quad p = C_1\bar{\sigma}\exp\left(-\frac{3}{2}\bar{\epsilon}\right)$

We have $\exp\left(-\frac{3}{2}\bar{\epsilon}\right) \approx 1$ for $\bar{\epsilon} \ll 1.0$,

and $\therefore p \approx C_1\bar{\sigma}$

But $p = \frac{2\bar{\sigma}t_0}{r_0}$ $\quad \therefore \quad C_1 = \frac{2t_0}{r_0}$

$$\therefore p = \frac{2t_0\bar{\sigma}}{r_0}\exp\left(-\frac{3}{2}\bar{\epsilon}\right)$$

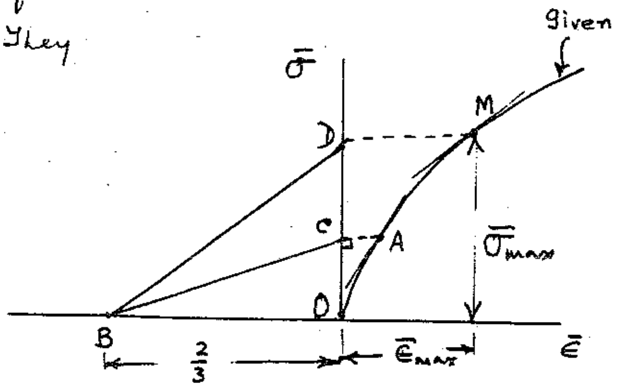
The max pressure occurs at $\bar{\sigma} = \sigma_{max}$, $\bar{\epsilon} = \epsilon_{max}$, as determined from the stress-strain curve.

$$p_{max} = \frac{2t_0\bar{\sigma}_{max}}{r_0}\exp\left(-\frac{3}{2}\bar{\epsilon}_{max}\right)$$

Determination of $\bar{\sigma}_{max}$ & $\bar{\epsilon}_{max}$:

To test point A, compare slope of tangent at A with slope of BC. They should be equal. For the curve shown, M is approximately the right point:

at M, $\frac{d\bar{\sigma}}{d\bar{\epsilon}} = \frac{3}{2}\bar{\sigma} \left(= \frac{OD}{OB}\right)$



9.17 Equation 9.26 $F = A\sigma$

Also $A = \dfrac{A_0 L_0}{L_0+\delta} = \dfrac{A_0}{1+\frac{\delta}{L_0}} = \dfrac{A_0}{1+\epsilon}$

$\therefore\ F = \dfrac{A_0\sigma}{1+\epsilon}$, $\dfrac{F}{A_0} = \dfrac{\sigma}{1+\epsilon}$

Fig. 9.12 is sketched alongside.

$\sigma_{max} = PQ$

$\dfrac{F_{max}}{A_0} = \dfrac{\sigma_{max}}{1+\epsilon}$

From geometry $\dfrac{OC}{QP} = \dfrac{1}{1+\epsilon}$

$\therefore OC = \dfrac{QP}{1+\epsilon} = \dfrac{\sigma_{max}}{1+\epsilon}$

$\therefore\ \dfrac{F_{max}}{A_0} = OC.$

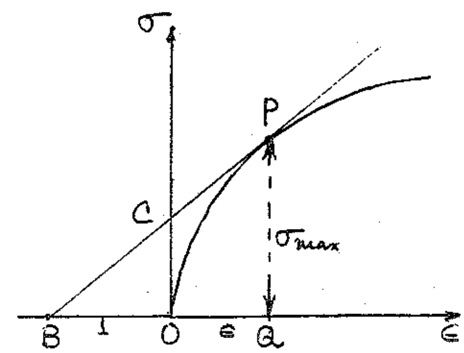
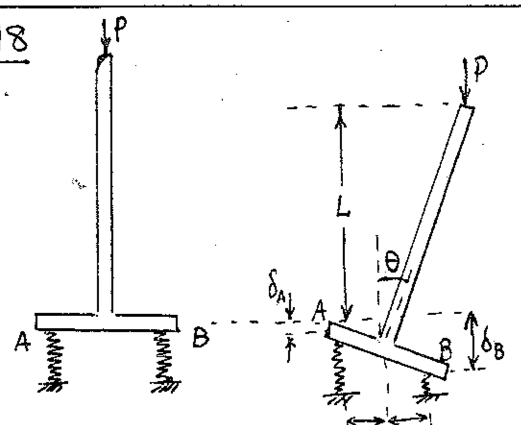
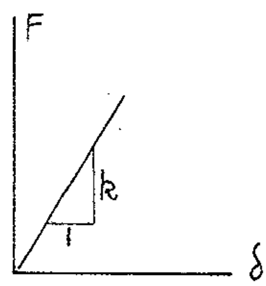


Fig 9.12

9.18





Equations 9.30 and 9.31 apply:

$F_A = \frac{P}{2}\left(1-\frac{L\theta}{c}\right)$ ——— (1)

$F_B = \frac{P}{2}\left(1+\frac{L\theta}{c}\right)$ ——— (2)

$\delta_B - \delta_A = 2c\theta$ ——— (3)

The force-deflection relations: $\left.\begin{aligned}F_A &= k\delta_A\\ F_B &= k\delta_B\end{aligned}\right\}$ $F_B - F_A = 2kc\theta$ [from (3)] → (4)

Substituting in (4) from (1) and (2), get

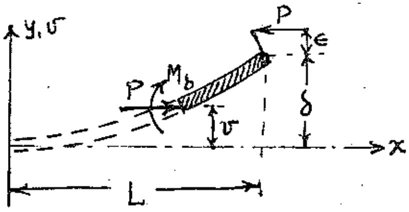
$$\theta\left(2kc - \frac{PL}{c}\right) = 0.$$

The non-trivial solution is $P_{cr} = \dfrac{2kc^2}{L}$, with arbitrary θ.

9.19

For the free-body shown,

$$M_b = P(\delta+\epsilon-v).$$



Equation 8.4 is $EI\dfrac{d^2v}{dx^2} = M_b$

$$\therefore \quad EI\frac{d^2v}{dx^2} = P(\delta+\epsilon-v) \quad \text{--- (1)}$$

Write $\phi \equiv -(\delta+\epsilon-v)$ ——(2)

Then (1) becomes

$$EI\frac{d^2\phi}{dx^2} + \frac{P}{EI}\phi = 0 \qquad \text{——— (3)}$$

The solution to (3) is

$$\phi = C_1 \sin x\sqrt{\frac{P}{EI}} + C_2 \cos x\sqrt{\frac{P}{EI}}$$

Substituting for ϕ,

$$v = \delta+\epsilon + C_1 \sin x\sqrt{\frac{P}{EI}} + C_2 \cos x\sqrt{\frac{P}{EI}}$$

Boundary Conditions: at $x=0$, $v = \dfrac{dv}{dx} = 0$; at $x=L$, $v=\delta$

These give $C_1 = 0$, $C_2 = -(\delta+\epsilon)$.

$$\therefore \quad v = (\delta+\epsilon)\left[1-\cos x\sqrt{\frac{P}{EI}}\right]$$

$$(v)_{x=L} = \delta = (\delta+\epsilon)\left[1-\cos L\sqrt{\frac{P}{EI}}\right]$$

This gives
$$\delta = \frac{\epsilon\left(1-\cos L\sqrt{\frac{P}{EI}}\right)}{\cos L\sqrt{\frac{P}{EI}}} = \epsilon\left(\sec L\sqrt{\frac{P}{EI}}-1\right).$$

9-20

FROM EQUILIBRIUM

$$(1) \qquad \frac{P}{2kL}(x+\epsilon) = x\left(1+\beta\frac{x^2}{L^2}\right)$$

DIFFERENTIATE W.R.T. x

$$\frac{d}{dx}\left(\frac{P}{2kL}\right)(x+\epsilon) + \frac{P}{2kL} = 1+3\beta\frac{x^2}{L^2}$$

AT POINT M, $\frac{d}{dx}\left(\frac{P}{2kL}\right) = 0$

$$(2) \qquad 1-\frac{P}{2kL} = -3\beta\frac{x^2}{L^2}$$

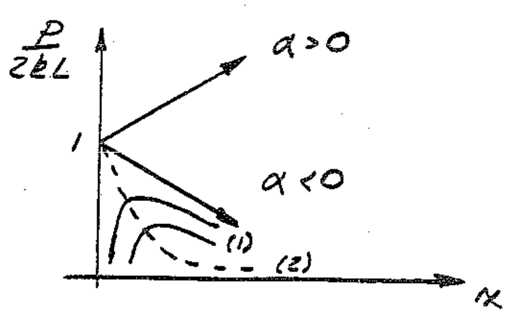
ELIMINATING $\frac{x}{L}$ BETWEEN (1) & (2)

$$\left(1-\frac{P}{2kL}\right)^{3/2} = \frac{3\sqrt{-3\beta}}{2}\,\frac{\epsilon}{L}\,\frac{P}{2kL} \qquad \text{AND} \begin{cases} P_{CRIT} = 2kL \\ P_{MAX} = P \end{cases}$$

9-21

FROM EQUILIBRIUM

$$\frac{P}{2kL}x = x\left(1+\alpha\frac{x}{L}\right)$$



WITH ECCENTRICITY

$$(1) \qquad \frac{P}{P_{CRIT}}(x+\epsilon) = x\left(1+\alpha\frac{x}{L}\right)$$

DIFFERENTIATE W. R. T. x

$$\frac{d}{dx}\left(\frac{P}{P_{CRIT}}\right)(x+\epsilon) + \frac{P}{P_{CRIT}} = 1+2\alpha\frac{x}{L}$$

BUT AT MAX. PT. $\frac{d}{dx}\left(\frac{P}{P_{CRIT}}\right) = 0$, $\left(1-\frac{P}{P_{CRIT}}\right) = -2\alpha\frac{x}{L}$ (2)

ELIMINATING x FROM (1) & (2)

$$\left(1-\frac{P}{P_{CRIT}}\right)^2 = -4\alpha\frac{\epsilon}{L}\frac{P}{P_{CRIT}}$$

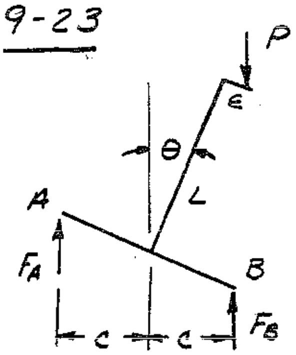
9-22

ASSUME α REMAINS FIXED, AS

θ GROWS FROM ZERO TO θ
P GROWS FROM P_0 TO P_d

THEN FROM (9.34)
$$P_d - P_0 = \theta\left[(k_c+k_t)\alpha - (k_c-k_t)c\right]$$
ALSO
$$P_d = \frac{2c^2}{L}\frac{2k_c k_t}{k_c+k_t} \quad (9.39)$$
$$P_0 = \frac{c}{L}\left[(k_c+k_t)c - (k_c-k_t)\alpha\right] \quad (9.35)$$
SOLVING FOR θ
$$\theta = \frac{P_d - P_0}{(k_c+k_t)\alpha - (k_c-k_t)c}$$
SUBSTITUTING FOR P_d & P_0
$$\theta = \frac{c}{L}\frac{k_c-k_t}{k_c+k_t}$$
WHICH IS INDEPENDENT OF $\frac{\alpha}{c}$

9-23



FROM EQUILIBRIUM
$$F_B + F_A = P$$
$$c(F_B - F_A) = P(\epsilon + L\theta)$$

AS P INCREASES, F_A, F_B, θ ALL INCREASE UNTIL $P = P_r$ WHEN F_A REACHES ITS MAXIMUM AND STARTS TO UNLOAD. FOR A DIFFERENTIAL INCREASE dP AT $P = P_r$, THE SYSTEM PIVIOTS $d\theta$ ABOUT POINT A SO THAT
$$dF_A = 0 \qquad dF_B = dP$$
$$d\delta_A = 0 \qquad d\delta_B = 2c\,d\theta$$

FOR THE SPRING AT B THE TANGENT MODULUS, k_t, IS
$$k_{tB} = \frac{dF_B}{d\delta_B}$$
DIFFERENTIATE THE MOMENT EQUILIBRIUM EQUATION
$$c(dF_B - dF_A) = dP(\epsilon + L\theta) + PL\,d\theta$$

SUBSTITUTING

$$c\, dF_B = dF_B\,(\epsilon + L\theta) + \frac{P_r L}{2c}\, d\delta_B$$

$$\frac{P_r L}{2c} = c\left(1 - \frac{\epsilon + L\theta}{c}\right)\frac{dF_B}{d\delta_B}$$

$$P_r = \frac{2c^2}{L}\, k_{t_B}\left(1 - \frac{\epsilon + L\theta}{c}\right)$$

BUT FROM (9.36) THE TANGENT MODULUS LOAD P_t IS

$$P_t = \frac{2c^2}{L}\, k_t$$

SO THAT

$$P_r < P_t$$

EVEN IF THE SAME MODULUS k_t IS USED. ACTUALLY k_t DEPENDS ON LOAD HISTORY. IN THE DEFINITION OF P_t IT IS ASSUMED THAT $F_A = F_B = P/2$ UP TO THE BIFURCATION AT P_t. IN THE ECCENTRICALLY LOADED COLUMN STUDIED HERE, F_B IS ALWAYS GREATER THAN F_A SO THAT $k_{t_B} \leq k_t$ FOR THE USUAL STRESS-STRAIN CURVE, CONCAVE DOWNWARD.